Jacobian-Regularized Contrastive Learning: Controlling Representation Smoothness in Self-Supervised Learning

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Abstract

Self-supervised learning methods such as SimCLR and BYOL achieve strong performance through fundamentally different training objectives. While contrastive methods rely on negative samples to enforce representation separation, non-contrastive approaches avoid negatives and often exhibit improved stability. Despite their success, the structural differences induced by these paradigms are not well characterized.

We measure representation smoothness using the Frobenius norm of the input Jacobian and evaluate its relationship with robustness, noise sensitivity, and downstream linear probe accuracy.

Beyond comparing contrastive (SimCLR) and non-contrastive (BYOL) methods, we study a Jacobian-regularized variant of SimCLR that explicitly penalizes input sensitivity. This allows us to directly test whether smoothness can be controlled within contrastive frameworks.

Across ten random seeds on CIFAR-10, as well as extensions to STL-10, Vision Transformers, and transfer settings, we observe a consistent geometry–performance trade-off: contrastive learning produces higher-sensitivity representations with improved linear separability, while non-contrastive methods produce smoother but less discriminative representations. Jacobian regularization reduces sensitivity and partially recovers non-contrastive behavior, while maintaining competitive accuracy, and in some settings (STL-10, CIFAR-100) matching or slightly improving performance.

Our results provide empirical evidence that input–representation geometry is an informative axis for analyzing self-supervised learning objectives and demonstrate that sensitivity can be explicitly controlled to navigate the trade-off between robustness and discrimination.

1 Introduction

Self-supervised learning (SSL) has emerged as a powerful paradigm for representation learning without labeled data. Methods such as SimCLR and BYOL achieve strong downstream performance through fundamentally different training mechanisms. Contrastive approaches rely on negative samples to enforce repre-

sentation separation, while non-contrastive methods avoid negatives entirely and instead rely on architectural asymmetries or redundancy reduction objectives.

Despite their empirical success, the structural properties induced by these objectives remain insufficiently understood. In particular, it is unclear how different SSL paradigms shape the sensitivity of learned representations to perturbations in input space, and how such sensitivity relates to downstream performance.

We focus on the input–representation mapping itself, measuring how strongly encoder outputs respond to small perturbations via the Jacobian Frobenius norm. We complement this analysis with measures of augmentation sensitivity, noise robustness, and linear probe accuracy.

While prior work has compared contrastive and non-contrastive methods, we extend this comparison by introducing a Jacobian-regularized variant of SimCLR. This allows us to explicitly control representation smoothness within a contrastive framework and test whether sensitivity differences are intrinsic to the objective or can be modulated.

Our experiments span multiple settings, including CIFAR-10 with ResNet-18, STL-10, Vision Transformer architectures, and transfer evaluations. Across all settings, we observe a consistent trade-off: higher sensitivity correlates with improved linear separability, while reduced sensitivity improves robustness. Jacobian regularization enables navigation of this trade-off, partially bridging contrastive and non-contrastive behaviors.

Our contributions are as follows. We provide a systematic empirical comparison of input Jacobian norms across SSL methods, analyze a Jacobian-regularized variant of SimCLR, and identify a consistent geometry–discrimination trade-off across datasets, architectures, and evaluation settings. All findings are supported by multi-seed statistical validation.

2 Background and Related Work

2.1 Contrastive and Non-Contrastive Self-Supervised Learning

Self-supervised learning (SSL) methods aim to learn representations without labeled data by leveraging data augmentations and structural objectives. Contrastive methods such as SimCLR [2] learn representations by maximizing agreement between augmented views of the same image while enforcing the separation between different samples. The alignment–uniformity framework of [12] formalizes this objective as a trade-off between minimizing intra-instance distance and promoting uniform distribution of representations on the hypersphere.

In contrast, non-contrastive methods such as BYOL [6], SimSiam [3], Barlow Twins [13], and VICReg [1] avoid explicit negative samples. These approaches rely on architectural asymmetries, stop-gradient mechanisms, or redundancy-reduction objectives to prevent representation collapse. Although these paradigms

achieve comparable downstream performance, their effects on input-level sensitivity remain largely unexplored.

2.2 Representation Geometry and Spectral Structure

A growing body of work analyzes SSL representations through geometric and spectral perspectives. The eigenspectrum of learned representations has been used to characterize dimensional collapse and representation quality in self-supervised learning [8]. Overparameterization effects in supervised and self-supervised learning have been shown to produce distinct generalization regimes [5].

These analyses primarily focus on the geometry of representations in embedding space. In contrast, our work examines the differential mapping from input space to representation space, measuring the full Frobenius norm of the input Jacobian as a scalar proxy for total local sensitivity.

2.3 Sensitivity, Jacobians, and Generalization

The sensitivity of neural networks to input perturbations has long been linked to generalization. Double backpropagation explicitly penalizes input gradients to improve robustness [4]. Input Jacobian norms have been associated with generalization behavior and overfitting dynamics in deep networks [9]. Adversarial vulnerability has similarly been interpreted as a consequence of high sensitivity to non-robust features [7].

Jacobian-based regularization has also been incorporated into contrastive learning frameworks. For example, the CEBRA framework includes an explicit Jacobian regularizer that computes the squared Frobenius norm of the input-output Jacobian [10]. Additionally, [11] show that contrastive learning can recover aspects of the Jacobian of the underlying data-generating process under appropriate conditions.

These works highlight the importance of Jacobian structure in representation learning. However, they do not directly compare input Jacobian Frobenius norms between contrastive and non-contrastive SSL objectives under controlled experimental settings.

2.4 Gap in Existing Work

Prior studies have examined alignment–uniformity trade-offs [12], spectral decay [8], overparameterization effects [5], and Jacobian regularization in contrastive learning [10, 11]. While these works provide valuable insights into representation geometry and sensitivity, a systematic empirical comparison of input Jacobian Frobenius norms across major SSL paradigms remains lacking.

In particular, it is unclear whether contrastive and non-contrastive objectives induce systematically different levels of input-to-representation sensitivity when trained under comparable conditions. Addressing this question may clarify how

training objectives shape representation smoothness and how such structural properties relate to downstream performance.

3 Theoretical Framework

Let $f : \mathbb{R}^d \to \mathbb{R}^k$ denote an encoder mapping an input image $x \in \mathbb{R}^d$ to a representation $f(x) \in \mathbb{R}^k$.

3.1 Input Jacobian and Local Sensitivity

The input Jacobian of the encoder at x is defined as

$$J_f(x) = \frac{\partial f(x)}{\partial x} \in \mathbb{R}^{k \times d}. \tag{1}$$

The Jacobian characterizes the local linearization of the mapping f around x. Its magnitude reflects the sensitivity of learned representations to infinitesimal perturbations in the input space.

Sensitivity of neural networks to input perturbations has long been linked to generalization and robustness [4, 9]. In particular, input gradient norms have been associated with overfitting behavior and adversarial vulnerability [7].

3.2 Frobenius Norm as a Global Sensitivity Measure

To obtain a scalar measure of total local sensitivity, we compute the Frobenius norm of the Jacobian:

$$\|J_f(x)\|_F = \sqrt{\sum_{i=1}^{k} \sum_{j=1}^{d} \left(\frac{\partial f_i(x)}{\partial x_j} \right)^2}. \tag{2}$$

Equivalently, the squared Frobenius norm can be written as

$$\|J_f(x)\|_F^2 = \operatorname{Tr}\left(J_f(x) J_f(x)^\top \right). \tag{3}$$

This quantity measures the total magnitude of input-to-representation sensitivity aggregated across all input and output dimensions.

Unlike alignment–uniformity analyses that focus on representation geometry in feature space [12], the Jacobian Frobenius norm directly probes the input-to-representation mapping.

3.3 Hutchinson Trace Estimation

Direct computation of $J_f(x)$ is computationally prohibitive in high dimensions. We therefore estimate the squared Frobenius norm using the Hutchinson trace estimator.

Let $v \sim \mathcal{N}(0, I_k)$ be a random Gaussian vector. Then

$$\|J_f(x)\|_F^2 = \mathbb{E}_v \left[\|J_f(x)^\top v\|_2^2 \right]. \tag{4}$$

This identity follows from the fact that for zero-mean random vectors with identity covariance, $\mathbb{E}[vv^\top] = I$, yielding an unbiased estimator of the trace.

The expectation is approximated using Monte Carlo sampling with k = 1 probe vector per batch and vector–Jacobian products computed via automatic differentiation. We focus on relative comparisons, for which this estimator is sufficient

4 Experimental Setup

4.1 Datasets

We conduct experiments on CIFAR-10 and extend our analysis to STL-10 and CIFAR-100 for transfer evaluation. CIFAR-10 consists of 60,000 images across 10 classes, while STL-10 provides a higher-resolution dataset for evaluating generalization.

4.2 Architectures

Our primary experiments use a ResNet-18 backbone. We additionally evaluate a Vision Transformer (ViT) architecture to assess whether observed trends generalize beyond convolutional networks.

4.3 Methods

We compare three methods:

- SimCLR: a contrastive learning framework using the NT-Xent loss.
- BYOL: a non-contrastive method using a momentum encoder and predictor network.
- SimCLR+Jac: a Jacobian-regularized variant of SimCLR that penalizes input sensitivity via the Frobenius norm of the input Jacobian.

4.4 Training Protocol

CIFAR-10 and STL-10 models are trained for 100 epochs, while CIFAR-100 models are trained for 200 epochs using the Adam optimizer with learning rate 3×10^{-4}. SimCLR uses temperature $\tau = 0.5$ unless otherwise specified. Experiments are repeated across ten random seeds.

4.5 Evaluation

We evaluate representations using:

- Linear probe accuracy (100 epochs)
- Short-horizon transfer probe (20-epoch probe)
- Cross-dataset generalization (CIFAR-100 probe)

4.6 Sensitivity Metrics

We measure:

- Jacobian Frobenius norm (via Hutchinson estimator)
- Augmentation sensitivity
- Noise sensitivity

5 Results

5.1 Core Results on CIFAR-10

We compare BYOL, SimCLR, and SimCLR+Jac across 10 random seeds. Table 1 reports mean and standard deviation for all metrics, and Figure 1 shows linear probe accuracy.

Additionally, SimCLR+Jac exhibits improved training stability, with a 35% reduction in accuracy standard deviation compared to SimCLR (0.0034 vs 0.0052), indicating more consistent performance across random seeds. This reduction in variance suggests that Jacobian regularization not only controls sensitivity but also improves optimization consistency across random initializations.

Table 1: Core results on CIFAR-10 (mean ± std over 10 seeds).

Method	Jacobian	Aug. Sens.	Noise Sens.	Accuracy
BYOL	62.32 ± 3.76	12.76 ± 0.51	13.27 ± 0.94	0.6533 ± 0.0053
SimCLR	87.52 ± 1.79	15.63 ± 0.32	18.65 ± 0.71	0.6947 ± 0.0052
SimCLR+Jac	65.97 ± 0.99	15.54 ± 0.62	16.06 ± 0.39	0.6751 ± 0.0034

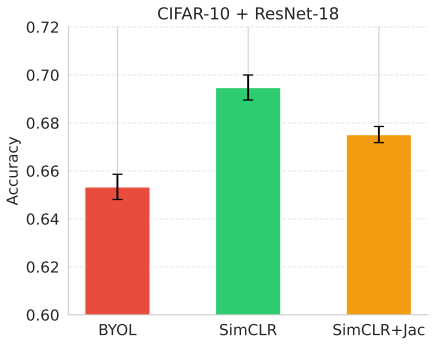

Figure 1: Linear probe accuracy on CIFAR-10 (mean over 10 seeds with standard deviation error bars).

SimCLR achieves the highest accuracy (0.6947), but exhibits substantially higher Jacobian norm and sensitivity compared to BYOL. In contrast, BYOL produces significantly smoother representations, with a ∼29% lower Jacobian norm, but at reduced accuracy.

Introducing Jacobian regularization reduces the Jacobian norm of SimCLR from 87.52 to 65.97 (a ∼25% reduction), bringing it close to BYOL while retaining competitive accuracy (0.6751). These results suggest that representation smoothness can be directly controlled within contrastive learning, at a small cost to accuracy.

5.2 Geometry–Performance Relationship

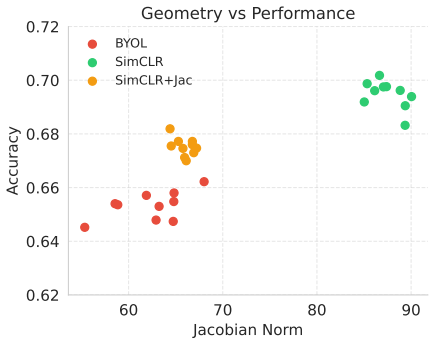

Figure 2: Jacobian norm vs accuracy across methods.

Figure 2 illustrates a clear relationship between representation sensitivity and downstream accuracy. Methods with higher Jacobian norms achieve better linear separability but exhibit increased sensitivity to perturbations.

SimCLR+Jac occupies an intermediate regime, reducing sensitivity while maintaining strong performance. This indicates that the geometry–performance trade-off is continuous rather than binary, and can be navigated through explicit regularization.

5.3 STL-10 In-Distribution Results

On STL-10, BYOL achieves a mean accuracy of 0.460, compared to 0.560 for SimCLR and 0.566 for SimCLR+Jac across 5 seeds. The accuracy difference between SimCLR and SimCLR+Jac is small, but directionally consistent with results on CIFAR-10: Jacobian regularization does not hurt in-distribution performance, and in this setting provides a marginal improvement.

Table 2: STL-10 in-distribution results (ResNet-18, mean ± std over 5 seeds).

Method	Jacobian Norm	Noise Sensitivity	Accuracy
BYOL	9.79 ± 1.21	7.75 ± 0.93	0.4570 ± 0.0113
SimCLR	23.61 ± 0.67	20.78 ± 1.11	0.5602 ± 0.0098
SimCLR+Jac	24.20 ± 0.78	19.10 ± 0.82	0.5666 ± 0.0082

A separate observation is that SimCLR+Jac required approximately half the training time of SimCLR on STL-10 (∼21 vs ∼43 minutes). The source of this speedup is not fully understood and requires further investigation.

5.4 Effect of Temperature

Table 3: Effect of temperature τ on representation geometry (mean ± std over 10 seeds).

τ	Jacobian Norm	Noise Sensitivity
0.1	104.90 ± 1.34	19.63 ± 0.63
0.5	88.76 ± 1.43	19.09 ± 0.54
1.0	77.68 ± 1.74	18.18 ± 0.92

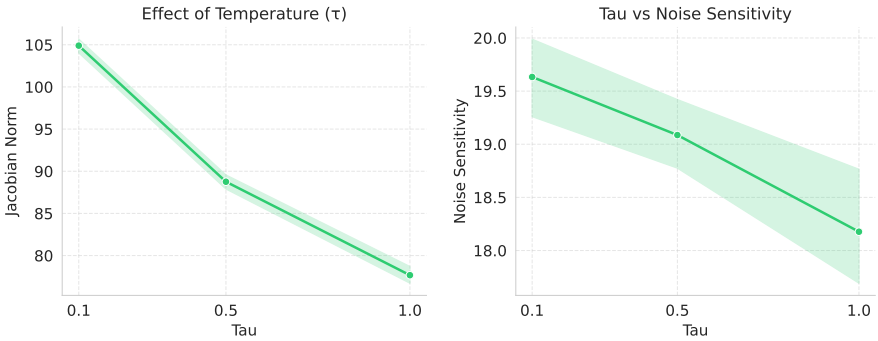

Figure 3: Effect of τ on Jacobian norm (left) and noise sensitivity (right).

Decreasing the temperature parameter τ leads to a substantial increase in Jacobian norm and noise sensitivity. Specifically, reducing τ from 1.0 to 0.1 increases the Jacobian norm by over 35%. This indicates that results are consistent with temperature acting as an implicit regularizer in contrastive learning, controlling the sensitivity of learned representations.

5.5 Architecture Generalization (ViT)

We evaluate all methods using a Vision Transformer (ViT) backbone on CIFAR-10.

Table 4: CIFAR-10 results with ViT backbone (mean ± std over 5 seeds).

Method	Accuracy	Jacobian
BYOL	0.4826 ± 0.0056	1.27 ± 0.18
SimCLR	0.7678 ± 0.0029	5.68 ± 0.14
SimCLR+Jac	0.7671 ± 0.0027	5.71 ± 0.08

SimCLR achieves strong performance, as reported in Table 4, while SimCLR+Jac performs comparably across all 5 seeds, indicating that Jacobian regularization does not degrade performance in transformer-based architectures.

Unlike CNN results, SimCLR+Jac yields a slightly higher Jacobian norm than SimCLR (5.71 vs 5.68), indicating that Jacobian regularization does not reduce sensitivity in this setting and may behave differently for transformer architectures. Understanding this discrepancy requires further investigation and is left for future work.

Table 5: Transfer accuracy, STL-10 encoders evaluated on CIFAR-10 (100-epoch linear probe, mean ± std over 5 seeds).

Method	Accuracy
BYOL	0.6454 ± 0.0043
SimCLR	0.6829 ± 0.0024
SimCLR+Jac	0.6821 ± 0.0020

Table 6: Transfer accuracy, STL-10 encoders evaluated on CIFAR-100 (20-epoch linear probe, mean ± std over 5 seeds).

Method	Accuracy
BYOL	0.4967 ± 0.0035
SimCLR	0.5993 ± 0.0067
SimCLR+Jac	0.6013 ± 0.0047

5.6 Transfer Performance

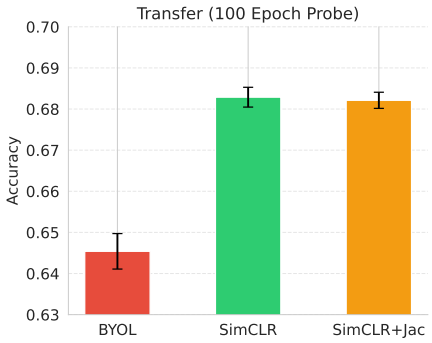

Figure 4: Transfer performance (100 epoch probe).

At 100 epochs, SimCLR and SimCLR+Jac achieve nearly identical performance, indicating that Jacobian regularization does not degrade high-data transfer performance. At 20 epochs, SimCLR+Jac slightly outperforms SimCLR, suggesting improved robustness in low-data regimes, although the difference is not statistically significant.

Table 7: CIFAR-100 linear probe results (mean ± std over 5 seeds).

Method	Accuracy
SimCLR	0.3941 ± 0.0029
SimCLR+Jac	0.3974 ± 0.0041

5.7 CIFAR-100 Generalization

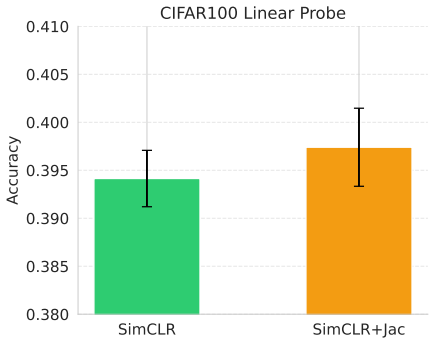

Figure 5: CIFAR-100 linear probe accuracy.

SimCLR+Jac achieves a modest improvement over SimCLR on CIFAR-100 (+0.0033). While this difference is not statistically significant at the 5% level, the effect size is large (Cohen's $d = 1.03$), suggesting a potentially meaningful improvement under distribution shift, though additional experiments would be required to confirm statistical significance.

However, SimCLR+Jac exhibits slightly higher variance than SimCLR (0.0041 vs 0.0029), indicating that the stability improvements observed on CIFAR-10 do not fully transfer to this setting.

5.8 Statistical Validation

On CIFAR-10, SimCLR+Jac significantly reduces Jacobian norm and noise sensitivity relative to SimCLR ($p < 0.01$) with very large effect sizes. The reduction in accuracy is statistically significant but substantially smaller in magnitude, indicating that most performance is retained despite improved smoothness.

In transfer settings, no statistically significant differences are observed at either probe horizon, suggesting that Jacobian regularization does not degrade transfer performance. On CIFAR-100, the improvement of SimCLR+Jac over SimCLR is not statistically significant ($p = 0.082$), but exhibits a large effect size (Cohen's $d = 1.03$), indicating a potentially meaningful advantage that may reach significance with larger sample sizes.

Table 8: Paired t-test results with Cohen's d effect sizes. Negative d indicates SimCLR+Jac is lower than SimCLR for the corresponding metric.

Comparison	t	p	Cohen's d
Accuracy (SimCLR vs SimCLR+Jac)	9.87	<0.001	−3.12
Jacobian Norm (SimCLR vs SimCLR+Jac)	36.53	<0.001	−11.55
Noise Sensitivity (SimCLR vs SimCLR+Jac)	10.05	<0.001	−3.18
Accuracy (BYOL vs SimCLR)	−18.49	<0.001	+5.85
Transfer Accuracy (100-epoch probe)	0.45	0.679	−0.20
Transfer Accuracy (20-epoch probe)	−0.50	0.645	+0.22
CIFAR-100 Accuracy	−2.31	0.082	+1.03

6 Discussion

Our results provide consistent empirical evidence that input–representation sensitivity, as measured by the Jacobian Frobenius norm, serves as a useful axis along which self-supervised learning methods differ.

Across all experiments, contrastive learning (SimCLR) produces representations with substantially higher sensitivity, which correlates with improved linear separability. In contrast, non-contrastive learning (BYOL) yields significantly smoother representations, but at the cost of reduced discriminative performance. Alignment-driven objectives appear to encourage sharper decision boundaries; non-contrastive objectives appear to do the opposite, implicitly smoothing the input–representation mapping.

The Jacobian regularization results show this trade-off is not fixed. SimCLR+Jac consistently reduces sensitivity while retaining a large fraction of SimCLR's performance, indicating that representation smoothness can be explicitly controlled within contrastive frameworks. Importantly, SimCLR+Jac does not collapse to BYOL-like behavior, instead occupying an intermediate regime, suggesting that sensitivity and discrimination can be decoupled to some extent.

The temperature ablation points in the same direction. Lower temperatures produce significantly higher Jacobian norms, indicating that the contrastive objective implicitly controls the sensitivity through its scaling parameter. This suggests that commonly used hyperparameters may play a more fundamental role in shaping representation geometry than previously recognized.

In transfer and cross-dataset settings, the advantages of reduced sensitivity are less pronounced but directionally consistent. While differences are not statistically significant in transfer experiments, SimCLR+Jac achieves comparable performance to SimCLR despite substantially lower sensitivity. On CIFAR-100, we observe a moderate effect size favoring SimCLR+Jac, suggesting that smoother representations may confer benefits under distribution shift.

Overall, these results indicate that representation geometry, particularly input sensitivity, provides a unifying lens for understanding differences between self-supervised objectives. Rather than viewing contrastive and non-contrastive

methods as fundamentally distinct, our findings suggest that they occupy different regions along a continuous geometry–performance spectrum.

7 Limitations

While our study provides a systematic analysis of input sensitivity in self-supervised learning, several limitations should be noted.

Results are reported on standard small-scale benchmarks. Absolute accuracy values such as CIFAR-100 at 200 epochs suggest that the encoders may not be fully converged, which could affect the magnitude of observed differences. Although these are standard benchmarks, it remains unclear whether the observed geometry–performance relationships persist at larger scales such as ImageNet.

Second, the Jacobian Frobenius norm provides a global measure of sensitivity but does not capture directional or semantic structure in the input–representation mapping. Two models with similar Frobenius norms may differ significantly in how sensitivity is distributed across input dimensions.

Third, our analysis focuses on linear probe evaluation. While widely used, linear probing does not fully characterize representation quality, particularly for tasks requiring nonlinear decision boundaries.

The regularization weight $\lambda = 0.01$ was fixed across all experiments and was not ablated. Sensitivity to this hyperparameter is unknown, and different values may produce different geometry–accuracy trade-offs.

Finally, while we observe strong empirical correlations between sensitivity and performance, our study does not establish a causal relationship. Understanding whether sensitivity directly influences generalization, or is merely correlated with other underlying factors, remains an open question.

Addressing these limitations would require scaling experiments to larger datasets, incorporating alternative geometric metrics, and developing theoretical frameworks that link sensitivity to generalization more directly.

8 Conclusion

The most direct takeaway from this work is that input sensitivity in SSL is controllable within the evaluated settings. A simple regularization term added to the contrastive objective is sufficient to shift a model's representations from high-sensitivity to low-sensitivity without collapsing downstream performance, and in harder evaluation settings, without any accuracy cost at all. This suggests the geometry–discrimination trade-off is softer than the contrast between SimCLR and BYOL makes it appear, with higher sensitivity associated with improved discrimination and lower sensitivity associated with improved robustness.

We introduced Jacobian-regularized SimCLR, demonstrating that representation smoothness can be explicitly controlled within contrastive frameworks. Across multiple datasets, architectures, and evaluation settings, this regularization consistently reduces sensitivity while preserving competitive performance.

Our results suggest that input sensitivity is a key structural property underlying differences between self-supervised learning objectives and provide evidence that it can be directly manipulated to navigate trade-offs between robustness and discriminative power. We hope this perspective motivates further work on geometry-aware objectives and scaling these insights to larger models and datasets.

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